

Open Source Physical Computing for Beginners

Physical computing is not as intimidating as it sounds. Enter the Arduino platform—a platform for physical computing for the beginners

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Just like chef Gusteau said in the movie *Ratatouille*, “Anyone can cook!” we could say, “Anyone can prototype!” Let’s begin by simplifying the term physical computing. It definitely sounds too techie to a newbie. According to Wikipedia, physical computing, in the broadest sense means building interactive physical systems by the use of software and hardware that can sense and respond to the analogue world.

In other words, we can make devices that use sensors as input and perform some action (physical or virtual) as output.

What is the Arduino platform?

Arduino (pronounced as *aar-do-we-know*) platform is a small and efficient AVR development board for hobbyists. It has been developed by Massimo Banzi and the Arduino team. AVR is the series of 8-bit RISC microcontrollers based on a modified Harvard microcontroller architecture. The chips used on the Arduino board include the ATmega AVR family—ATmega8, ATmega168, ATmega328 and the ATmega1280.

Since the Arduino platform is Open Source, there have been many clones, and many of these clones use

other processors too apart from the ones listed above. The Arduino board comes in many sizes for a varied number of applications, the most common being the Arduino Duemilanove. The Diecimila is the older version of the Duemilanove. Other form factors include Arduino LilyPad (for wearable computing; it can be sewn onto/into clothes), Arduino Nano (for use with breadboards), Arduino Mega (a more powerful variant of the Duemilanove), Arduino Pro (a cheaper version of the Duemilanove meant for advanced users), Arduino Pro mini, Arduino Fio and Arduino BT (for wireless applications; BT uses Bluetooth) and Arduino Serial (unlike the other boards, it uses a serial/RS-232 interface instead of the USB for hook-up to the computer).

The Duemilanove board is suitable for starters. It is easy to use since it’s USB-based.

The fundamentals

The first thing is to get hold of the hardware. In India, you can get the board from feemo or rhydoLABZ. The cheapest way to assemble an Arduino is to use the breadboard. Here, a blank Arduino PCB with serial COM port is used and the components are soldered onto it. The parts list is available on the official Arduino site with the schematic. It is fairly easy to assemble.

The Arduino board with serial COM port requires power (unlike the Duemilanove where

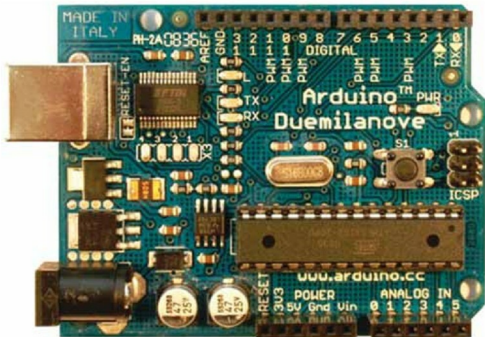


Fig. 1: The Arduino board